sunlight is only about 10 percent what it is on Earth. Still, we could imagine vast fields of solar panels covering the surfaces of inhabited asteroids and converting sunlight into electricity. Photovoltaic technology is routinely used in Earth-orbiting spacecraft, and is in increasing use on the surface of the Earth today. But while that might be enough to warm and light the homes of these descendants, it does not seem adequate to change asteroid orbits.

For that, Williamson proposed using anti-matter. Anti-matter is just like ordinary matter, with one significant difference. Consider hydrogen: An ordinary hydrogen atom consists of a positively charged proton on the inside and a negatively charged electron on the outside. An atom of anti-hydrogen consists of a negatively charged proton on the inside and a positively charged electron (also called a positron) on the outside. The protons, whatever the sign of their charges, have the same mass; and the electrons, whatever the sign of *their* charges, have the same mass. Particles with opposite charges attract. A hydrogen atom and an anti-hydrogen atom are both stable, because in both cases the positive and negative electrical charges precisely balance.

Anti-matter is not some hypothetical construct from the perfervid musings of science fiction writers or theoretical physicists. Anti-matter exists. Physicists make it in nuclear accelerators; it can be found in high-energy cosmic rays. So why don't we hear more about it? Why has no one held up a lump of anti-matter for our inspection? Because matter and anti-matter, when brought into contact, violently annihilate each other, disappearing in an intense burst of gamma rays. We cannot tell whether something is made of matter or anti-matter just by looking at it. The spectroscopic properties of, for example, hydrogen and anti-hydrogen are identical.

Albert Einstein's answer to the question of why we see only matter and not anti-matter was, "Matter won"—by which he meant that in our sector of the Universe at least, after almost all the matter and anti-matter interacted and annihilated each other long ago, there was some of what we call ordinary matter left over.* As far as we can tell today, from gamma ray astronomy and other means, the